MACHINE LEARNING • PROBLEM-BASED LEARNING

Online Game Price Prediction Using Machine Learning

Course: Machine Learning (PBL)

Saish Nehe — 2124UMLM1051

Nikhil Jadhav — 2124UMLM1096

Omprakash Wagh — 2124UMLM1073

Tushar Borawake — 2124UMLM1037

PROJECT SCOPE

Problem and objective

Estimate the observed Steam game price from game characteristics.

Use a labelled public dataset and compare regression algorithms.

Provide an interactive Streamlit interface for educational demonstration.

The target is the dataset snapshot price in euros, not a live or future quote.

Dataset and features

Source: Steam Games - UOC Project on Zenodo (DOI: 10.5281/zenodo.15121027).

Record describes 3,537 games collected in March 2025 from the Spanish-language Steam store.

Target: price (EUR). Inputs: genres, launch_year, reviews, general_reviews.

Exclude discount and original_price from predictors to reduce target leakage.

Preprocessing and EDA

Inspect data types, missing values, duplicates, and scraping-error flags.

Parse European price formats and convert review counts to numeric values.

Extract launch year and impute missing values within model pipelines.

Create price distribution, genre summary, and reviews-versus-price charts.

Models and methodology

Median baseline and Ridge Regression.

Decision Tree Regressor, Random Forest Regressor, and Extra Trees Regressor.

Use an 80/20 train-test split with a fixed random seed.

Select the model by lowest mean 5-fold cross-validation MAE on training data.

Evaluation

MAE: average absolute prediction error in euros.

RMSE: penalises larger prediction errors more strongly.

R^2: describes explained variation in the held-out test set.

Insert actual results from reports/model_comparison.csv and reports/test_metrics.json after running training.

No performance score is claimed before experiments are executed.

DEPLOYMENT

Application

Streamlit form accepts genre, launch year, review count, and review summary.

Loads the trained pipeline and metadata produced by the training script.

Displays predicted price and measured test metrics.

Not a live Steam checker; users must check the official store for current regional prices.

Plan and rubric

1. Setup environment and download data. 2. Inspect and clean. 3. Perform EDA.

4. Engineer features and compare models. 5. Evaluate the selected model on test data.

6. Build and test the Streamlit app. 7. Finish report, notebook, presentation, and README.

Covers all 10 evaluation criteria: problem, data, cleaning, EDA, features, models, evaluation, insights, deployment, documentation.

FINAL SUBMISSION

Deliverables

Executable Jupyter Notebook and Python source code.

Dataset source link, data dictionary, preprocessing and EDA visualisations.

Model comparison, actual evaluation results, insights, and limitations.

Working application or deployment demonstration.

Project report, presentation, organised code, and README.